

**1-Q-Write a program in python to accept marks in three subjects (out of 100) from the user. calculate the total marks and average**

```
subject1 = float(input("Enter marks for subject 1 (out of 100): "))
subject2 = float(input("Enter marks for subject 2 (out of 100): "))
subject3 = float(input("Enter marks for subject 3 (out of 100): "))
total_marks = subject1 + subject2 + subject3
average = total_marks / 3
print("Total Marks:", total_marks)
print("Average Marks: ",average)
```

**2- write a python program to calculate his gross salary, Rohan basic salary is input through the keyboard. His dearness allowance is 40% of basic salary, and house rent allowance is 20% of basic salary**

```
basic_salary = float(input("Enter Your basic salary: "))
    # Calculating da(40% of basic salary)
da = 0.40 * basic_salary
    # Calculating hra (20% of basic salary)
hra = 0.20 * basic_salary
    # Calculating gross salary
gross_salary = basic_salary + da + hra
    # Displaying the calculated allowances and gross salary
print("Basic Salary:",basic_salary)
print("Dearness Allowance (40% of basic):",da)
print("House Rent Allowance (20% of basic):",hra)
print("Gross Salary:",gross_salary)
```

**3- Write a Python program to reverse three digits number without using loop.**

```
number = int(input("Enter a three digit number: "))
    #hundreds, tens, and ones place digits
hundred = number // 100
tens = (number // 10) % 10
ones = number % 10

rev_num = ones * 100 + tens * 10 + hundred

print("Reversed number:",rev_num)
```

**4-Write an Python program to calculate amount after deducting 10% of discount. Enter the sale amount from keyboard.**

```
sale_amt = float(input("Enter the sale amount: "))
discount = sale_amt * (10/100)#discount
final_amount = sale_amt - discount
print("The amount after 10% discount is:",final_amount)
```

**5-Write an Python program to calculate the profit. Enter the sales price and cost price through keyboard.**

```
s_price = float(input("Enter the sales price: "))
cost_price = float(input("Enter the cost price: "))
profit = s_price - cost_price
print("The profit is:" profit)
```

**6-Python Program to Check Whether a Student is Pass or Fail(marks are greater than or equal to 35)**

```
subject1 = int(input('Enter the Mark:'))
subject2 = int(input('Enter the Mark:'))
subject3 = int(input('Enter the Mark:'))
subject4 = int(input('Enter the Mark:'))
subject5 = int(input('Enter the Mark:'))
if(subject1>=35 and subject2>=35 and subject3>=35 and subject4>=35 and
subject5>=35):
    print('Pass')
else:
    print('Fail')
```

**7-Python Program to Calculate Grade of a Student**

**If the average is greater than 90, "Grade: A" is printed.**

**If the average is in between 80 and 90, "Grade: B" is printed.**

**If the average is in between 70 and 80, "Grade: C" is printed.**

**If the average is in between 60 and 70, "Grade: D" is printed.**

**If the average is anything below 60, "Grade: F" is printed.**

```
sub1=int(input("Enter marks of the first subject: "))
sub2=int(input("Enter marks of the second subject: "))
sub3=int(input("Enter marks of the third subject: "))
sub4=int(input("Enter marks of the fourth subject: "))
sub5=int(input("Enter marks of the fifth subject: "))
avg=(sub1+sub2+sub3+sub4+sub4)/5
if(avg>=90):
    print("Grade: A")
elif(avg>=80&avg<90):
    print("Grade: B")
elif(avg>=70&avg<80):
    print("Grade: C")
elif(avg>=60&avg<70):
    print("Grade: D")
else:
    print("Grade: F")
```

## 8- Simple Calculator without functions

```
# Taking input from the user
print("Select operation:")
print("1. Add")
print("2. Subtract")
print("3. Multiply")
print("4. Divide")
choice = input("Enter choice (1/2/3/4): ")
# Getting the two numbers from the user
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
# Performing operations based on the user's choice
if choice == '1':
    result = num1 + num2
    print(num1,"+",num2,"=",result)
elif choice == '2':
    result = num1 - num2
    print(num1,"-",num2,"=",result)
elif choice == '3':
    result = num1 * num2
    print(num1,"*",num2,"=",result)
elif choice == '4':
    if num2 != 0:
        result = num1 / num2
        print(num1,"/",num2,"=",result)
    else:
        print("Cannot divide by zero!")
else:
    print("Invalid input! Please select a valid input choice.")
```

## 9-# Print digits of a number in reverse order

```
num = int(input("Enter a number: "))
while num > 0:
    digit = num % 10 # Get the last digit
    print(digit, end=" ")
    num = num // 10 # Remove the last digit
```

## 10-WAPP to input any number from the keyboard and check it number is palindrome or not

```
num=int(input("Enter any number:"))
temp=num
rev=0
while(num>0):
    dig=num%10
    rev=rev*10+dig
    num=num//10
if(temp==rev):
    print(temp," is Palindrome.")
else:
    print(temp," is not Palindrome.")
```

### 11- Python program to check if the number is an Armstrong number or not

```
num=int(input("Enter any number:"))
temp=num
sum=0
while(num>0):
    dig=num%10
    sum=sum+dig**3
    num=num//10
if(temp==sum):
    print(temp," is Armstrong.")
else:
    print(temp," is not Armstrong.")
```

### 12-Print Star pattern

```
*
* *
* * *
* * * *
* * * * *

no=int(input("Enter no of row's:"))
for i in range(1,no+1):
    for j in range(1,i+1):
        print("*",end=" ")
    print()
```

### 13- Print pattern

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

```
no=int(input("Enter no of row's:"))
for i in range(1,no+1):
    for j in range(1,i+1):
        print(i,end=" ")
    print()
```

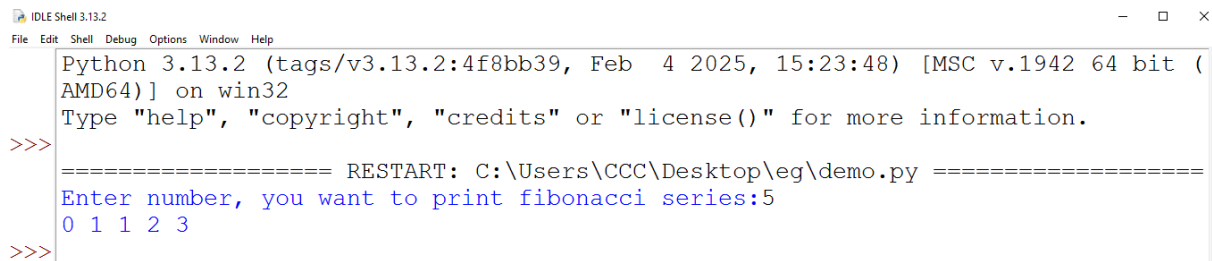
### 14- Print pattern

```
no=int(input("Enter no of row's:"))
for i in range(1,no+1):
    print(" "*(no-i),end=" ")
    print(""*i)
```

```
===== RESTART: C:\Users\CCC\Desktop\eg\demo.py =:
Enter no of row's:5
    *
   **
  ***
 ****
*****
>>>|
```

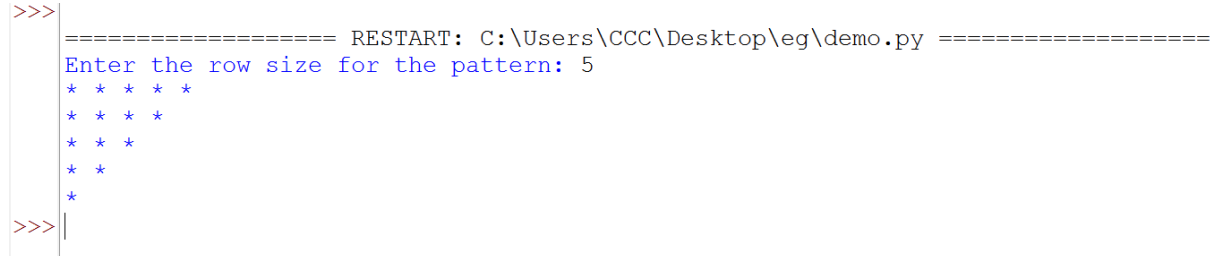
### 15-Write a python program to print Fibonacci series

```
n=int(input("Enter number, you want to print fibonacci series:"))
first=0
second=1
if n <=0:
    print("Incorrect input, Please enter any other number:")
elif n == 1:
    print(first,end=" ")
else:
    for i in range(n):
        print(first,end=" ")
        temp=first
        first=second
        second=temp+first
```



```
IDLE Shell 3.13.2
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:\Users\CCC\Desktop\eg\demo.py =====
Enter number, you want to print fibonacci series:5
0 1 1 2 3
>>>
```

### 16- Print pattern( Inverted Right-Angled Triangle in Python)



```
>>>
===== RESTART: C:\Users\CCC\Desktop\eg\demo.py =====
Enter the row size for the pattern: 5
* * * * *
* * * *
* * * *
* * *
* *
*
>>>
```

```
rows = int(input("Enter the row size for the pattern: "))
for i in range(rows, 0, -1): # Outer loop for rows in reverse
    for j in range(1, i + 1): # Inner loop for columns
        print("*", end=" ") # Print star
    print()
```

### 17- Pyramid Pattern in Python

```
rows = int(input("Enter the row size for the pattern: "))
for i in range(1, rows + 1): # Outer loop for rows
    for j in range(rows - i): # Inner loop for spaces
        print(" ", end=" ")
    for k in range(1, 2 * i): # Inner loop for stars
        print("*", end=" ")
    print()
```

```

===== RESTART: C:\Users\CCC\Desktop\eg\demo.py =====
Enter the row size for the pattern: 5
      *
     * * *
    * * * * *
   * * * * * * *
  * * * * * * * *
 >>>

```

### 18- Alphabet Right-Angled Triangle Pattern in Python

```
rows = int(input("Enter the row size for the pattern: "))
```

```
for i in range(1, rows + 1): # Outer loop for rows
```

```
    for j in range(1, i + 1): # Inner loop for columns
```

```
        print(chr(64 + j), end=" ") # Print alphabets
```

```
    print()
```

```

===== RESTART: C:\Users\CCC\Desktop\eg\demo.py =====
Enter the row size for the pattern: 5
A
A B
A B C
A B C D
A B C D E
>>>

```

### 19-Sequential Search in Python

```
def seq_search(arr, target):
```

```
    for i in range(len(arr)):
```

```
        if arr[i] == target:
```

```
            return i
```

```
            # Target found at index i
```

```
    return -1
```

```
    # Target not found
```

```
numbers = [20, 25, 5, 55, 35, 18]
```

```
target = 55
```

```
result = seq_search(numbers, target)
```

```
if result != -1:
```

```
    print(f'Element found at index {result}')
```

```
else:
```

```
    print("Element not found")
```

```

===== RESTART: C:/Users/CCC/Desktop/eg/sequential_search.py
Element found at index 3

```

## 20-Binary search

```
def binary_search(arr, target):
    low = 0
    high = len(arr) - 1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == target:
            return mid # Found the target
        elif arr[mid] < target:
            low = mid + 1 # Search right half
        else:
            high = mid - 1 # Search left half

    return -1 # Not found

numbers = [2, 4, 6, 8, 10, 12, 14] # Must be sorted
target = 10

result = binary_search(numbers, target)

if result != -1:
    print(f"Target value Found at index position: {result}")
else:
    print("Target value Not found:")
```

## Step by step Execution

### Iteration 1:

- `low = 0, high = 6`
- `mid = (0 + 6) // 2 = 3`
- `arr[mid] = arr[3] = 8`
- `8 < 10` → Target is in the **right half**
- Update: `low = mid + 1 = 4`

### Iteration 2:

- `low = 4, high = 6`
- `mid = (4 + 6) // 2 = 5`
- `arr[mid] = arr[5] = 12`
- `12 > 10` → Target is in the **left half**
- Update: `high = mid - 1 = 4`

### Iteration 3:

- `low = 4, high = 4`
- `mid = (4 + 4) // 2 = 4`
- `arr[mid] = arr[4] = 10`

- Target found! Return index 4

**21-calculate series  $x + x^3/3! + x^5/5! + x^7/7! + \dots$  or**

$$x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \dots$$

**Step 1: Observe the pattern**

- Powers of  $x$ : odd powers only  $\rightarrow x^1, x^3, x^5, x^7, \dots$
- Denominators: factorials of odd numbers  $\rightarrow 1!, 3!, 5!, 7!, \dots$
- All signs are positive

**#The symbol! means: factorial**

```
import math
x = float(input("Enter the value of x: "))
n = int(input("Enter the number of terms: "))
result = 0
# Generate n terms
power = 1                                # start with x^1
for i in range(n):
    term = x**power / math.factorial(power)
    result += term
    power += 2                            # next odd power
print(f"Sum of the series up to {n} terms is: {result}")
```

**output-**

```
Enter the value of x: 2
Enter the number of terms: 3
Sum of the series up to 3 terms is: 3.5999999999999996
```

**22- Python Program to Calculate Multiplication Using Repeated Addition:**

```
a = int(input("Enter first number (a): "))
b = int(input("Enter second number (b): "))
result = 0
```

```
for i in range(b):
    result += a
print(f"{a} × {b} = {result}")
```

**output-**

```
Enter first number (a): 6
Enter second number (b): 7
6 × 7 = 42
```

**23- Write a program to calculate the telephone bill on the basis of following Condition:**

- For first 200 calls free
- For next 100 calls Rs. 0.80/call
- For next 100 calls Rs. 1.80/call
- For next calls Rs. 2.30/call

```
calls = int(input("Enter the number of calls made: "))
```



```
bill = 0.0

if calls <= 200:
    bill = 0
elif calls <= 300:
    bill = (calls - 200) * 0.80
elif calls <= 400:
    bill = 100 * 0.80 + (calls - 300) * 1.80
else:
    bill = 100 * 0.80 + 100 * 1.80 + (calls - 400) * 2.30

print(f"Total telephone bill is: Rs. {bill:.2f}")
```

**output-**

```
Enter the number of calls: 450
Total telephone bill is: Rs. 375.00
```